📦 IPv4 Header (10.25.2026)

> **Quick Summary:** The IPv4 header is the metadata attached to every data packet. It tells routers where the packet came from, where it is going, and how to handle it.
> 🎥 *Source: Jeremy's IT Lab CCNA Course - Day 10*

🗺️ How to Read the IPv4 Header Table (Bit-Map)

The header is structured as a grid. Think of it as a spreadsheet where data is processed in **32-bit chunks**.

- **Top Columns:** Represent individual **Bits** (0 to 31).
- **Left Rows:** Represent **Octets (Bytes)**. Because 1 byte = 8 bits, each 32-bit row equals **4 Octets (Bytes)**.

💡 **The IHL Formula (Minimum vs. Actual Length):**

- **Minimum Header Length:** 20 Bytes (the physical reality of all mandatory fields combined).
- **Minimum IHL Value:** 5 (The IHL field counts in 4-byte chunks, not individual bytes. `20 bytes ÷ 4 = 5`).
- **Maximum IHL Value:** 15 (`15 × 4 = 60 bytes`, which is the maximum possible header size with Options).
- **The Rule:** Always multiply the IHL number by 4 to get the actual header length in bytes.

IPv4 header format

Offset	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Version (4)				IHL				DSCP						ECN		Total Length															
4	32	Identification																Flags			Fragment Offset											
8	64	Time to Live								Protocol								Header Checksum															
12	96	Source address																															
16	128	Destination address																															
20	160	(Options) (if IHL > 5)																															
⋮	⋮																																
56	448																																

🏷️ Row-by-Row Field Breakdown

🔹 Row 1: Basics & Size (Octet 0 / Bits 0-31)

- **Version (4 bits):** Specifies IPv4. *Fixed binary value:* `0100` *(decimal 4).*
- **IHL (4 bits):** Header length. *Min: 5 (20 bytes). Max: 15 (60 bytes).*
- **DSCP & ECN (8 bits):** Quality of Service (QoS) and congestion notification.
- **Total Length (16 bits):** Size of the entire packet (Header + Data). *Max value: 65,535 bytes.*

🔹 Row 2: Packet Fragmentation (Octet 4 / Bits 32-63)

Used when a packet exceeds the Maximum Transmission Unit (MTU) of a link.

- **Identification (16 bits):** Unique ID to group fragments together.
- **Flags (3 bits):** Indicates if the packet can be fragmented. *(Note: The **More Fragments bit** is set to* `1` *on all fragments except the last one).*
- **Fragment Offset (13 bits):** Tells the receiver where this fragment belongs in the original packet.

🔹 Row 3: Routing & Integrity (Octet 8 / Bits 64-95)

- **Time to Live / TTL (8 bits):** Prevents infinite loops. Decrements by 1 at every router hop. *If it hits* `0`*, the packet is dropped.*
- **Protocol (8 bits):** Identifies the L4 protocol inside. *(See Protocol Cheat Sheet below).*
- **Header Checksum (16 bits):** Error-checks the *header only*. (Errors in the actual *data* are detected by TCP/UDP).

🔹 Row 4 & 5: Addressing (Octet 12 & 16 / Bits 96-159)

- **Source Address (32 bits):** Sender's IP.
- **Destination Address (32 bits):** Receiver's IP.

🔹 Row 6+: Options (Octet 20+ / Bits 160+)

- **Options:** *Variable in length.* Rarely used. If present, it increases the header size beyond 20 bytes (up to a max of 60 bytes).

🛣️ Protocol Cheat Sheet (L4PDU)

Decimal Value	Protocol
1	ICMP (Ping)
6	TCP
17	UDP
89	OSPF (Routing)

🧠 Knowledge Check (Quiz)

1. **What is the fixed binary value of the first field of an IPv4 header?**
   - a) 0010
   - b) 0110
   - c) 0001
   - **d) 0100 — ANSWER** *(This is the "Version" field, representing decimal 4).*
2. **Which field will cause the packet to be dropped if it has a value of 0?**
   - **a) TTL — ANSWER** *(If a router sees a TTL of 0, it discards the packet to prevent loops).*
   - b) DSCP
   - c) IHL
   - d) ECN
3. **How are errors in an IPv4 packet's encapsulated data detected?**
   - a) The IPv4 Header Checksum field checks for errors.
   - **b) The encapsulated protocol (TCP, UDP) checks for errors. — ANSWER** *(The IPv4 Header Checksum only checks the header, not the payload).*
   - c) Errors in the encapsulated data cannot be detected.
4. **Which field of an IPv4 header is variable in length?**
   - **a) Options — ANSWER** *(The IHL field is fixed at 4 bits. The Options field itself varies in size, from 0 to 40 bytes).*
   - b) Header Checksum
   - c) Total Length
   - d) IHL
5. **Which bit will be set to 1 on all IPv4 packet fragments except the last fragment?**
   - a) Fragment Offset bit
   - **b) More Fragments bit — ANSWER** *(It tells the receiver "wait, there's more data coming").*
   - c) Don't Fragment bit
   - d) Packet Fragment bit